

ZHUO LI

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🎓 EDUCATION

Institute of Computing Technology, Chinese Academy of Sciences (Recommended for Admission), *M.S.* in Computer Application Technology 2019 – 2022

Laboratory: Key Lab of Intelligent Information Processing, VIPL Group

Advisor: Shuqiang Jiang, Professor, National Distinguished Young Scholar

Northwestern Polytechnical University, *B.S.* in Computer Science and Technology
(Comprehensive Ranking: 5/234) 2015 – 2019

⚙️ SKILLS

- **Programming & Frameworks:** Proficient in Python, C++; Familiar with deep learning frameworks such as PyTorch, TensorFlow.
- **Core Technologies:** Deep understanding and extensive deployment experience in Video Generation (Diffusion Models, RGBA), Multimodal LLMs (Autoregressive models, VLM), 3D Human Motion Capture & Generation, Embodied AI & Physics Simulation (Reinforcement Imitation Learning, Humanoid Robot Control), 3D Vision, and Image Retrieval.

👥 WORK & RESEARCH EXPERIENCE

ByteDance · Senior Research Engineer Nov. 2025 – Present

- **Focus:** Building intelligent creation capabilities for video generation, including 3D generation, animation generation, and transparent-channel (RGBA) video generation.
- **UniRGBA: Transparent-Channel Video Generation Framework** (First author)
 - **Method:** Proposed UniRGBA, a unified framework extending a causal video VAE to 4-channel RGBA via zero-initialized channel expansion and symmetric selective layer unfreezing. Applied standard DoRA to a pretrained DiT, reusing the unified 4-channel latent without alpha-specific architectural changes.
 - **Achievements:** Outperformed baseline methods on RGBA reconstruction and generation while preserving the underlying RGB prior and fidelity. Under review at **NeurIPS 2026**.

Tencent · Research Engineer Jul. 2022 – Nov. 2025

- **Monocular/Multi-view 3D Human Motion Capture (Offline & Real-time)** (First author)
 - **Method & Achievements:** To mitigate depth ambiguity in **offline motion capture**, proposed an iterative technique based on human motion priors and camera parameter learning, improving ground estimation accuracy and substantially reducing foot sliding and human leaning, achieving **SOTA**. The **real-time motion capture** system runs at 45fps on-device (CPU) with notable precision improvement over baseline.
- **Multimodal, Multitask Motion Comprehension and Generation (M^3 GPT)**
 - **Method & Achievements:** Compressed motion and music into discrete tokens to establish a unified representation space. Introduced auxiliary tasks (music-to-text, text-to-dance) for modality alignment. Constructed an end-to-end **Autoregressive** LLM framework addressing audio-rhythm alignment and motion coherence. Accepted by **NeurIPS 2024**.
- **Any-Modality Conditional Motion Generation Framework (AnyMo)** (First author)
 - **Method & Achievements:** Constructed **OmniHuMo**, a large-scale multimodal motion dataset. Proposed AnyMo combining a Residual FSQ-based tokenizer and a scalable masked modeling transformer for high-

fidelity, controllable motion synthesis under arbitrary modality combinations. Under review at **NeurIPS 2026**.

- **Animation Generation & Physical Plausibility Repair (Morph)** (First author)
 - **Method & Achievements:** Proposed a motion-free physical repair framework using reinforcement imitation learning and a physics engine. Effectively repaired foot sliding, floating, and ground penetration, achieving **SOTA**. Also applied to humanoid robot imitation learning. Accepted by **ICCV 2025**.
- **Video Effects Generation Pipeline**
 - **Method & Achievements:** Developed a Wan 2.1-based effects generation pipeline with a complete development toolbox to address slow inference and high development complexity. Supported body-, music-, and voice-driven human video generation effects.

Meituan · Research Intern

Dec. 2020 – Jul. 2021

- **Fine-grained Image Retrieval** (First author)
 - **Method & Achievements:** Proposed a novel **PNP Loss** to improve Average Precision optimization redundancy, accelerating training and enhancing retrieval. Achieved **SOTA** on three public datasets. Merged into the open-source library PML (5.6k+ stars). Co-organized the ICCV FoodAI Workshop.
- **Domain Adaptation for Stereo Matching**
 - **Method & Achievements:** Identified that smooth L1 loss and soft argmin suppress generalization by causing multi-modal distributions in unseen domains. Proposed **Uncertainty-regularized Minimization** and **Anisotropic Soft Argmin** to constrain the disparity distribution. Won the **Champion** of the Argoverse Stereo Competition (**CVPR 2021**). Accepted by Pattern Recognition 2025.

PUBLICATIONS

1. UniRGBA: Unifying RGB and Alpha in Pretrained Video Diffusion Models.
Zhuo Li[†], Lei Xu*, Yongyi Jin*, Aoyu Wang. *Under review at Advances in Neural Information Processing Systems (NeurIPS 2026)*. (**Co-first & Corresponding author, CCF-A**).
2. AnyMo: Scaling Any-Modality Conditional Motion Generation with Masked Modeling.
Yiheng Li*, **Zhuo Li*** (**Project leader**), Ruibing Hou, Yingjie Chen, Hong Chang, Hao Liu, Shiguang Shan. *Under review at Advances in Neural Information Processing Systems (NeurIPS 2026)*. (**CCF-A**).
3. Morph: A Motion-free Physics Optimization Framework for Human Motion Generation.
Zhuo Li*, Mingshuang Luo*, Ruibing Hou, Xin Zhao, Hao Liu, Hong Chang, Zimo Liu, Chen Li. *International Conference on Computer Vision (ICCV)*, 2025. (**Co-first author, CCF-A**).
4. Rethinking the Optimization of Average Precision: Only Penalizing Negative Instances before Positive Ones is Enough.
Zhuo Li, Weiqing Min, Jiajun Song, Yaohui Zhu, Liping Kang, Xiaoming Wei, Xiaolin Wei, Shuqiang Jiang. *AAAI Conference on Artificial Intelligence*, 2022. (**CCF-A**).
5. CMD: Constraining Multimodal Distribution for Domain Adaptation in Stereo Matching.
Zhelun Shen*, **Zhuo Li***, Chenming Wu, Zhibo Rao, Lina Liu, Yuchao Dai, Liangjun Zhang. *Pattern Recognition*, 2025. (**SCI Q1**).
6. Toward Food Image Retrieval via Generalization-oriented Sampling and Loss Function Design.
Jiajun Song*, **Zhuo Li***, Weiqing Min, and Shuqiang Jiang. *ACM Transactions on Multimedia Computing, Communications, and Applications (ToMM)*, 2023. (**Co-first author, CCF-B**).
7. M^3 GPT: An Advanced Multimodal, Multitask Framework for Motion Comprehension and Generation.
Mingshuang Luo, Ruibing Hou, **Zhuo Li**, Hong Chang, Zimo Liu, Yaowei Wang, Shiguang Shan. *The 38th Conference on Neural Information Processing Systems (NeurIPS 2024)*. (**CCF-A**).
8. Ingredient-Guided Region Discovery and Relationship Modeling for Food Category-Ingredient Prediction.
Zhiling Wang, Weiqing Min, **Zhuo Li**, Liping Kang, Xiaoming Wei, Xiaolin Wei, and Shuqiang Jiang. *IEEE Transactions on Image Processing (TIP)*, 2022. (**CCF-A**).
9. A Two-Stage Triplet Network Training Framework for Image Retrieval.
Weiqing Min, Shuhuan Mei, **Zhuo Li** and Shuqiang Jiang. *IEEE Transactions on Multimedia (TMM)*,

2020. (SCI Q1).

10. A Noise-robust Locality Transformer for Fine-grained Food Image Retrieval.
Jiajun Song, Weiqing Min, Yuxin Liu, **Zhuo Li**, Shuqiang Jiang, and Yong Rui. *IEEE 5th International Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2022.

♡ HONORS AND AWARDS

<i>National Scholarship</i>	Oct. 2018
<i>Champion, Argoverse Stereo Competition (CVPR 2021)</i>	Jun. 2021
<i>4th Place, eBay eProduct Visual Search Challenge (CVPR 2021)</i>	May 2021
<i>Outstanding Graduate, Northwestern Polytechnical University</i>	Jul. 2019
<i>2nd Place in Northwest Region, 1st Prize at National Level, iCAN Contest</i>	Dec. 2018
<i>1st Prize at International Level, International Underwater Robot Competition</i>	Jul. 2018

⚙️ ADDITIONAL INFORMATION

- Academic Services: Reviewer for NeurIPS, AAAI, CVPR, ICCV, ECCV, and other journals and conferences.